SUPERSTORE SALES PERFORMANCE ANALYSIS

Executive Summary | PostgreSQL and Microsoft Power BI

1. Executive Summary

This project analysed Superstore sales data using PostgreSQL and Microsoft Power BI to evaluate overall business performance, category profitability, regional sales, monthly trends, and customer and product performance. The validated dataset contained 10,194 records, with 5,111 orders and 804 customers.

The business recorded total sales of 2,326,534.52 and total profit of 292,297.57, giving an overall profit margin of approximately 12.56%. Technology led in sales and profit, the West region had the highest regional sales, and Furniture's low profit margin emerged as an area for further investigation.

2. Overall Business Performance

Key Performance Indicator	Result
Total Sales	**2,326,534.52**
Total Profit	**292,297.57**
Total Orders	**5,111**
Total Customers	**804**
Overall Profit Margin	**12.56%**

Finding: The business generated positive overall profit. However, performance varied by category, so overall profitability should be considered alongside product- and category-level results.

3. Product Category Performance

Category	Sales	Profit	Profit Margin
Technology	839,893.35	146,544.09	17.45%
Office Supplies	731,893.23	126,023.54	17.22%
Furniture	754,747.94	19,729.94	2.61%

Technology recorded the highest sales and profit. Office Supplies also performed strongly, with a profit margin of approximately 17.22%. Furniture generated substantial sales but had a much lower margin of 2.61%, about 14.84 percentage points below Technology.

Business implication: Investigate Furniture's product costs, discounts, shipping expenses, and individual product profitability. The current figures identify a concern but do not establish its cause.

4. Regional Sales Performance

Region	Total Sales	Performance
West	739,813.71	Highest
South	391,721.90	Lowest

The West region recorded the highest sales, while the South region recorded the lowest among the four regions. The difference between them was 348,091.81. Investigate demand, order volumes, product availability, and regional profitability before deciding on actions.

5. Monthly Sales Performance

Month	Total Sales	Performance
November	352,666.01	Highest
February	59,751.26	Lowest

Sales varied considerably between the months analysed. November was the strongest month and February the weakest. Review order volumes, product mix, promotions, and other records to understand the reasons for this difference.

6. Customer and Product Performance

Measure	Finding	Value
Top customer by sales	Sinmula	25,400.07
Top product by sales	Canon imageCLASS 2200 Advanced Copier	61,599.83

The dashboard also examines Consumer, Corporate, and Home Office segments. Reviewing segment purchasing patterns, repeat purchases, and product profitability can help identify opportunities for retention and growth. High sales alone do not necessarily indicate high profit.

7. Recommendations

1. Investigate Furniture profitability

Review product costs, discounting, shipping expenses, and individual product margins. Furniture's 2.61% margin is substantially below the margins of Technology and Office Supplies.

2. Build on Technology's performance

Identify the products contributing most to Technology's sales and profit, then monitor their margins and customer demand before extending successful practices to other categories.

3. Investigate South region sales

Explore customer demand, order volumes, product availability, sales mix, and regional profitability to identify opportunities for improvement.

4. Understand monthly sales variations

Investigate the factors behind November's highest sales and February's lowest sales. Compare order volumes, product mix, promotions, and other relevant records.

5. Strengthen customer and product analysis

Review purchasing patterns among high-value customers and assess product profitability. High sales alone do not necessarily mean high profit.

6. Monitor performance regularly

Use the Power BI dashboard to track sales, profit, orders, customers, category performance, regional results, and monthly trends.

8. Conclusion

The analysis demonstrates how PostgreSQL and Power BI can turn sales records into actionable business insights. Technology was the strongest category by sales and profit, the West region led regional sales, and November had the highest monthly sales. Furniture requires further profitability investigation because its margin was only 2.61%.

The main recommendation is to investigate underperforming products and categories while learning from stronger performers. These findings indicate where further analysis is useful; they do not by themselves prove the causes of performance differences.

9. Methodology and Limitations

The work used Excel for initial preparation, PostgreSQL for data validation and SQL-based analysis, and Power BI Desktop with DAX for KPI calculations and interactive visualisation. The report summarises the results discussed from the existing dashboard. Further cost, discount, and product-level analysis may be needed to explain the causes of differences in profitability.